

Appendix: Inventory of Existing IVB Simulation Designs

Task 10 static audit

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1 Scope and reproducibility

This appendix is a static audit of simulations already present in the repository. It does not run, source, or modify a Monte Carlo script. The companion builder, `build_simulation_design_inventory.R`, reads only source files and existing artifacts, verifies declared paths, records MD5 hashes, and writes the inventory and manifest. Equations and parameter statements labelled as verified are transcribed from the versioned scripts; parameter grids retained in timing/grid CSVs are referenced rather than re-created. Any unlinked legacy script or runtime-configurable repetition count is explicitly marked as a gap.

The inventory uses a deliberately narrow evidentiary rule: a result is *directly verified* only when its source script and an existing result artifact both exist. It does not certify that a CSV was regenerated in this audit, nor does it infer a causal interpretation from a coefficient shift alone.

2 Coverage matrix

Table 1: Required coverage and the directly mapped inventory rows.

Coverage requirement	Mapped inventory IDs	Count
Dual-role dynamics	diag_heterogeneous; diag_ivb_sign; diag_sensitivity; diag_shortT; diag_timeFE; dual_role_8models; dual_role_asymmetry; dual_role_base; dual_role_firewall; dual_role_varyT; dual_role_varyT_8models; mechC_adl	12
Firewall comparison	callaway_controls; diag_sensitivity; diag_shortT; direct_carryover; direct_feedback; dual_role_8models; dual_role_firewall; dual_role_varyT; dual_role_varyT_8models; feedback_Y_to_D; feedback_carryover; mechC_adl; nl_carryover; nl_collider; nl_interact; persistent_unobserved; staggered_posttreat	17
Mediator over-control	jensen_twoperiod; overcontrol_contemp; overcontrol_full	3
Nonlinearity	callaway_controls; nl_carryover; nl_collider; nl_interact; overcontrol_contemp; overcontrol_full	6
Feedback	diag_phi; direct_feedback; feedback_Y_to_D; feedback_carryover	4
Carryover	direct_carryover; feedback_carryover; nl_carryover	3

Table 1: Required coverage and the directly mapped inventory rows. (*continued*)

Coverage requirement	Mapped inventory IDs	Count
Staggered binary treatment	staggered_posttreat; v4_mechC	2
Measurement error	v4_mechD	1
Level-dependent trends	callaway_controls	1
Contemporaneous confounding	jensen_twoperiod; persistent_unobserved	2

3 Design inventory

Table 2: Reader-facing overview of existing simulation DGPs and families.

Inventory ID	Simulation family	Estimand	Design	Coverage tags	Status
callaway-controls	Callaway-style nonlinear control relationship and level-dependent trends	beta=1 under stated DGP; diagnostic of specification robustness.	N=timing CSV; T=source; burn-in=source; reps=500	callaway style trends; nonlinearity; firewall	verified existing artifacts
diag-heterogeneous	diagnostic: heterogeneous treatment effects	mean beta / estimator target under heterogeneity.	N=source; T=source; burn-in=source; reps=200	heterogeneity; dual role	verified existing artifacts
diag-ivb-sign	diagnostic: IVB sign	beta=1 and sign of specification shift.	N=source; T=source; burn-in=source; reps=200	dual role	verified existing artifacts
diag-phi	diagnostic: feedback mechanism	beta=1 conditional on history.	N=source; T=source; burn-in=source; reps=300	feedback	verified existing artifacts
diag-sensitivity	diagnostic: parameter sensitivity	beta=1.	N=source; T=source; burn-in=source; reps=200	dual role; firewall	verified existing artifacts
diag-shortT	diagnostic: short-T / Nickell	beta=1; finite-T estimator behavior.	N=source; T=10;30; burn-in=source; reps=200	dual role; firewall	verified existing artifacts
diag-timeFE	diagnostic: time FE in DGP	beta=1.	N=source; T=source; burn-in=source; reps=200	dual role; time effects	verified existing artifacts
direct-carryover	linear direct carryover	contemporaneous beta=1; models may target misspecified effects when omitting D_lag.	N=timing CSV; T=source; burn-in=source; reps=500	carryover; firewall	verified existing artifacts
direct-feedback	lagged outcome-to-treatment feedback	beta=1 under stated history adjustment.	N=timing CSV; T=source; burn-in=source; reps=500	feedback; firewall	verified existing artifacts
dual-role-8models	dual-role Z full 2x2x2 estimator grid	beta=1.	N=100; T=30; burn-in=100; reps=500	dual role; firewall	verified existing artifacts
dual-role-asymmetry	dual-role causal-channel asymmetry	beta=1.	N=source; T=source; burn-in=source; reps=500	dual role	verified existing artifacts
dual-role-base	dual-role Z baseline	beta=1.	N=100; T=30; burn-in=100; reps=200	dual role	verified existing artifacts
dual-role-firewall	dual-role firewall sensitivity	beta=1.	N=source; T=source; burn-in=source; reps=500	dual role; firewall	verified existing artifacts
dual-role-varyT	dual-role finite-T variation	beta=1.	N=source; T=varies; burn-in=source; reps=200	dual role; firewall	declared output missing
dual-role-varyT-8models	dual-role finite-T full estimator grid	beta=1.	N=source; T=varies; burn-in=source; reps=500	dual role; firewall	verified existing artifacts
feedback-Y-to-D	feedback Y_lag to D with extended ADL	beta=1 conditional on recorded history.	N=timing CSV; T=source; burn-in=source; reps=500	feedback; firewall	verified existing artifacts
feedback-carryover	combined feedback and carryover	beta=1; misspecified models can target composite effects.	N=timing CSV; T=source; burn-in=source; reps=500	feedback; carryover; firewall	verified existing artifacts
jensen-twoperiod	Jensen-calibrated two-period DiD	ATT_total=tau_direct+y_zm*zn	N=10,000; T=2; burn-in=not applicable; reps=500 default (CLI override)	mediator overcontrol; contemporaneous unobserved confounding	verified existing artifacts

Table 2: Reader-facing overview of existing simulation DGPs and families. (*continued*)

Inventory ID	Simulation family	Estimand	Design	Coverage tags	Status
mechC-adl	dual-role / firewall ADL battery	direct $\beta=1$ under the stated DGP.	N=timing CSV; T=timing CSV; burn-in=source parameters; reps=500	dual role; firewall	verified existing artifacts
nl-carryover	nonlinear carryover	direct $\beta=1$ and model-specific estimands under carryover.	N=timing CSV; T=source; burn-in=source; reps=500	nonlinearity; carryover; firewall	verified existing artifacts
nl-collider	nonlinear collider / firewall	$\beta=1$.	N=timing CSV; T=10;30; burn-in=source; reps=500	nonlinearity; firewall	verified existing artifacts
nl-interact	nonlinear D-by-history interaction	$\beta=1$.	N=timing CSV; T=source; burn-in=source; reps=500	nonlinearity; firewall	verified existing artifacts
overcontrol-contemp	contemporaneous pure-mediator over-control	direct $\beta=1$; linear total effect $\beta+\theta \Delta_D$.	N=100; T=30; burn-in=100; reps=500 default (environment override)	mediator overcontrol; nonlinearity	verified existing artifacts
overcontrol-full	mediator over-control with nonlinear D-to-Z	linear total= $\beta+\theta \Delta_D$; direct= β ; nonlinear total is model-specific.	N=100; T=30; burn-in=100; reps=500	mediator overcontrol; nonlinearity	verified existing artifacts
persistent-unobserved	persistent contemporaneous unobserved confounding	$\beta=1$; oracle models condition on U.	N=timing CSV; T=source; burn-in=source; reps=500	contemporaneous unobserved confounding; firewall	verified existing artifacts
staggered-posttreat	absorbing staggered treatment with post-treatment lag	$\beta=1$ contemporaneous structural coefficient, not necessarily a generic event-study ATT.	N=timing CSV; T=source; burn-in=source; reps=500	staggered binary treatment; posttreatment lag; firewall	verified existing artifacts
v1-legacy-completa	v1 legacy exploratory bundle	varies by section.	N=mixed; T=mixed; burn-in=not stated; reps=500 in documented blocks	legacy	legacy no persisted result
v1-twfe	v1 baseline FWL/TWFE grid	β and FWL specification shift $\beta_{\text{long}}-\beta_{\text{short}}$.	N=source/result CSV; T=source/result CSV; burn-in=none stated; reps=500 per scenario	collider	verified existing artifacts
v4-mechA	v4 mechanism A: between/within D-to-Z	β and FWL shift.	N=200; T=30; burn-in=none; reps=500	collider	verified existing artifacts
v4-mechB	v4 mechanism B: between/within Y-to-Z	β and FWL shift.	N=200; T=30; burn-in=none; reps=500	collider	verified existing artifacts
v4-mechC	v4 mechanism C: staggered binary treatment	β and FWL shift.	N=200; T=30; burn-in=none; reps=500	collider; staggered binary treatment	verified existing artifacts
v4-mechD	v4 mechanism D: measurement error in Z	β and attenuation of FWL shift.	N=200; T=30; burn-in=none; reps=500	collider; measurement error	verified existing artifacts

The canonical inventory CSV retains equations, parameter grids, fixed effects, estimator details, source scripts, result paths, and notes for each row. Table 2 selects the fields needed to identify the design, target, scale, coverage, and audit status without reducing the CSV record.

4 Artifact provenance

Table 3: Static provenance summary for declared existing artifacts.

Path	MD5	Bytes	Static validation
simulations/ dynamics/ results/ sim_callaway_controls_results.csv	77b7a7e7844bcfad687a098402cc745d	41,625	PASS
simulations/ dynamics/ results/ sim_callaway_controls_raw.csv	171209a0803c4207d0c8c2d97f3d43be	5,348,512	PASS
simulations/ dynamics/ results/ sim_callaway_controls_timing.csv	e56347b9d61b6578c0305b06d876b7b6	2,877	PASS
simulations/ diagnostics/ results/ diag_heterogeneous_results.csv	70c492338d919956484c40c3036ae4db	980	PASS

Table 3: Static provenance summary for declared existing artifacts. (*continued*)

Path	MD5	Bytes	Static validation
simulations/ diagnostics/ results/ diag_ivb_sign_results.csv	cdf4bb75758f15a663daf5c63fd290e3	2,636	PASS
simulations/ diagnostics/ results/ diag_phi_bias_mechanism_results.csv	96e3b8a99f4f65d579aecf30f2ea7fba	556	PASS
simulations/ diagnostics/ results/ diag_sensitivity_params_results.csv	d71624f097a82e4e813c7ee894737037	2,275	PASS
simulations/ diagnostics/ results/ diag_short_T_results.csv	10832d9d32a0b1ea1a11c304891ac984	1,193	PASS
simulations/ diagnostics/ results/ diag_time_fe_results.csv	6915dd27317ec946d5390958fbf66939	1,866	PASS
simulations/ dynamics/ results/ sim_direct_carryover_results.csv	d1fa5a41586d2f8e5b992e4da32dc996	5,202	PASS
simulations/ dynamics/ results/ sim_direct_carryover_raw.csv	91f7d48b203250b5bb484a807527b7f1	711,514	PASS
simulations/ dynamics/ results/ sim_direct_carryover_timing.csv	df540a4c75b083cf69a98ebf05f21760	164	PASS
simulations/ dynamics/ results/ sim_direct_feedback_results.csv	1cd3033b6e35cf88c12d81a821e9ab85	3,154	PASS
simulations/ dynamics/ results/ sim_direct_feedback_raw.csv	d23a192cc9ed2634ca6a6ee4a25ec01f	344,325	PASS
simulations/ dynamics/ results/ sim_direct_feedback_timing.csv	e907fa69675c5033593e6b0a93b5726e	158	PASS
simulations/ dual_role_z/ results/ sim_dual_role_z_8models_results.csv	eb5ac53a9dd902b35764d40de1b28629	5,041	PASS
simulations/ dual_role_z/ results/ sim_dual_role_z_8models_raw.csv	25cca514b6ec2c33b409f0c94b648742	742,015	PASS
simulations/ dual_role_z/ results/ sim_dual_role_z_asymmetry_results.csv	c70f7fee90dde42f0f4c7e44f923a0c5	3,393	PASS
simulations/ dual_role_z/ results/ sim_dual_role_z_results.csv	b77f0509364c86788224fd6c5da30aae	3,972	PASS
simulations/ dual_role_z/ results/ sim_dual_role_z_raw.csv	9e04225b560f985750286ac7794091ca	321,391	PASS
simulations/ dual_role_z/ results/ sim_dual_role_z_firewall_results.csv	3a5a2bc3ac6d9abeff4197f9ceb19cbb	2,092	PASS
simulations/ dual_role_z/ results/ sim_dual_role_z_varyT_8models_results.csv	4914d306c2ba0d5f4f7e0131e624c4bb	3,431	PASS
simulations/ dynamics/ results/ sim_feedback_Y_to_D_results.csv	1cc3085c3f6885322c594c17b8dd7e64	6,795	PASS
simulations/ dynamics/ results/ sim_feedback_Y_to_D_raw.csv	1511e591f477dd51eb937a6d79506975	904,659	PASS
simulations/ dynamics/ results/ sim_feedback_Y_to_D_timing.csv	db39c6457203d672fa928017be68077f	65	PASS
simulations/ dynamics/ results/ sim_feedback_carryover_results.csv	fc9bfa98cd8e2fff2ede9d3cd79b2f44	4,130	PASS
simulations/ dynamics/ results/ sim_feedback_carryover_raw.csv	ffb5099f92e9e48911f1c91837bea1ff	469,999	PASS
simulations/ dynamics/ results/ sim_feedback_carryover_timing.csv	9ef0aa9056ec7f54aecdbcb64e7976626	170	PASS
simulations/ dynamics/ results/ sim_jensen_two_period_summary.csv	05b6961ce53319b442c4dfb42f6d8b1e	1,578	PASS
simulations/ dynamics/ results/ sim_jensen_two_period_raw.csv	211e6fbd9d0f396b5b6d0794b696b91	954,442	PASS
simulations/ dynamics/ results/ sim_jensen_two_period_decomp.csv	c5920edb0167065bf293eb6c82d7a286	1,236	PASS
simulations/ dynamics/ results/ sim_jensen_two_period_scenarios.csv	37443c176cf62a6456d0d8abe58e06fa	993	PASS
simulations/ v4_mechanisms/ results/ sim_mechC_adl_results.csv	039b09403a568bb4f9fb8bf290935059	41,977	PASS
simulations/ v4_mechanisms/ results/ sim_mechC_adl_raw.csv	e37935d7650b9d3d4e04e1b3c09057ca	6,346,856	PASS
simulations/ v4_mechanisms/ results/ sim_mechC_adl_timing.csv	c51e858d3a69a19a39cc7b2e0392ccf9	65	PASS
simulations/ nonlinearity/ results/ sim_nl_carryover_results.csv	f4b50985d3d11960dfd8c8a73e0e0ef1	9,818	PASS
simulations/ nonlinearity/ results/ sim_nl_carryover_raw.csv	a6c6a2ffa520f862bfc94cfa742275f4	1,149,275	PASS
simulations/ nonlinearity/ results/ sim_nl_carryover_timing.csv	faa3636f05c4115bfab0d87a0127af6d	708	PASS

Table 3: Static provenance summary for declared existing artifacts. (*continued*)

Path	MD5	Bytes	Static validation
simulations/ nonlinearity/ results/ sim_nl_collider_results.csv	effdf1df03d76b36f1cb0b5850373bed	58,965	PASS
simulations/ nonlinearity/ results/ sim_nl_collider_raw.csv	a4d990e1c4b9ccb5c4a4e3a8f90114d2	7,206,890	PASS
simulations/ nonlinearity/ results/ sim_nl_collider_timing.csv	7cd3ff8c0de40e0b951fdec1bc1f7d06	2,738	PASS
simulations/ nonlinearity/ results/ sim_nl_interact_results.csv	5693affc63d054d158e8cccb1cbbd18c	7,598	PASS
simulations/ nonlinearity/ results/ sim_nl_interact_raw.csv	62962188ab95fc2e0211565080577054	831,750	PASS
simulations/ nonlinearity/ results/ sim_nl_interact_timing.csv	28b66dd03707d10a47e603b6550a7bdd	292	PASS
simulations/ nonlinearity/ results/ sim_overcontrol_contemporaneous_results.csv	c426434b77997072b095a3d2abe0d282	15,433	PASS
simulations/ nonlinearity/ results/ sim_overcontrol_contemporaneous_raw.csv	4d118a1ebcef2b10cf3d2b4e96af74ef	3,670,455	PASS
simulations/ nonlinearity/ results/ sim_overcontrol_contemporaneous_timing.csv	b477293f6c5ba69d0e553d2aedcfb50b	110	PASS
simulations/ overcontrol/ results/ sim_overcontrol_results.csv	69480d070563983c0a9766478bff6302	32,026	PASS
simulations/ overcontrol/ results/ sim_overcontrol_raw.csv	def0176159f884a8cdfaddc51c8f29ae	3,899,487	PASS
simulations/ overcontrol/ results/ sim_overcontrol_grid.csv	b460dee64b0f278739f21139f9944527	2,708	PASS
simulations/ dynamics/ results/ sim_persistent_confounder_results.csv	92fe8a5c65711503d6241ccf54d7c1f6	44,342	PASS
simulations/ dynamics/ results/ sim_persistent_confounder_raw.csv	4ea07050167bf643fbed660ade434c9f	4,636,755	PASS
simulations/ dynamics/ results/ sim_persistent_confounder_timing.csv	184552a53a7424a7416d4d214a1ef5c4	842	PASS
simulations/ dynamics/ results/ sim_staggered_posttreat_results.csv	b9de7cb94f0a1f9460968e50786ed3f7	13,135	PASS
simulations/ dynamics/ results/ sim_staggered_posttreat_raw.csv	3f06af5e193c549f9f5549d76a325674	2,078,787	PASS
simulations/ dynamics/ results/ sim_staggered_posttreat_timing.csv	51c472cf144272524793c5134382525e	635	PASS
simulations/ v1_twfe/ sim_ivb_twfe_results.csv	2ac1f369ab903b6e55e35c25d41c3c15	146,951	PASS
simulations/ v4_mechanisms/ results/ sim_ivb_twfe_v4_mechA.csv	65aaf3f6bfeb3e41a32c5158a4b4ba29	32,532	PASS
simulations/ v4_mechanisms/ results/ sim_ivb_twfe_v4_mechB.csv	9348dfd74eecddfa8b1d902ebb4b6bdd	21,803	PASS
simulations/ v4_mechanisms/ results/ sim_ivb_twfe_v4_mechC.csv	333575ebc0d1150d48a289b078145cf5	4,892	PASS
simulations/ v4_mechanisms/ results/ sim_ivb_twfe_v4_mechD.csv	0ab671c8231567193abfdb835cf35ba0	4,898	PASS

5 Full CSV headers

Table 4 preserves the complete header record from the manifest. Headers are shown in a separate two-column longtable so that individual field names wrap rather than competing with the provenance fields in Table 3.

Table 4: Complete CSV headers recorded in the provenance manifest.

Artifact path	Full CSV header
simulations/ dynamics/ results/ sim_callaway_controls_results.csv	mechanism, nl_type, nl_str, lambda, delta_H, delta_Y, rho_Z, gamma_nl, twfe_s_mean, twfe_s_bias, twfe_s_mcse, twfe_s_rmse, twfe_l_mean, twfe_l_bias, twfe_l_mcse, twfe_l_rmse, adl_Ylag_mean, adl_Ylag_bias, adl_Ylag_mcse, adl_Ylag_rmse, adl_full_mean, adl_full_bias, adl_full_mcse, adl_full_rmse, adl_Dlag_mean, adl_Dlag_bias, adl_Dlag_mcse, adl_Dlag_rmse, adl_DYlag_mean, adl_DYlag_bias, adl_DYlag_mcse, adl_DYlag_rmse, adl_DZlag_mean, adl_DZlag_bias, adl_DZlag_mcse, adl_DZlag_rmse, adl_all_mean, adl_all_bias, adl_all_mcse, adl_all_rmse, adl_all_nofe_mean, adl_all_nofe_bias, adl_all_nofe_mcse, adl_all_nofe_rmse, n_sims, n_discarded
simulations/ dynamics/ results/ sim_callaway_controls_raw.csv	twfe_s.D, twfe_l.D, adl_Ylag.D, adl_full.D, adl_Dlag.D, adl_DYlag.D, adl_DZlag.D, adl_all.D, adl_all_nofe.D, sim, mechanism, nl_type, nl_str, gamma_nl, lambda, delta_H, delta_Y, rho_Z

Table 4: Complete CSV headers recorded in the provenance manifest. (continued)

Artifact path	Full CSV header
simulations/ dynamics/ results/ sim_callaway_controls_timing.csv	mechanism, nl_type, nl_str, lambda, delta_H, delta_Y, rho_Z, gamma_nl, n_discarded, total_s
simulations/ diagnostics/ results/ diag_heterogeneous_results.csv	sigma_beta, scenario, phi, beta2, twfe_s_bias, adl_full_bias, adl_all_bias, adl_full_rmse, adl_all_rmse
simulations/ diagnostics/ results/ diag_ivb_sign_results.csv	gamma_D, delta_Y, twfe_s, twfe_l, adl_s, adl_l, IVB_TWFE, IVB_ADL, net_TWFE, net_ADL
simulations/ diagnostics/ results/ diag_phi_bias_mechanism_results.csv	phi, rho_Y, adl_fe_bias, adl_nofe_bias, r2_D_Ylag
simulations/ diagnostics/ results/ diag_sensitivity_params_results.csv	rho_Y, gamma_D, gamma_Y, adl_full_bias, adl_all_bias, adl_full_rmse, adl_all_rmse, dominates
simulations/ diagnostics/ results/ diag_short_T_results.csv	TT, label, phi, beta2, twfe_l, adl_full, adl_all, adl_nofe, adl_all_rmse
simulations/ diagnostics/ results/ diag_time_fe_results.csv	label, phi, beta2, sigma_delta, twfe_s, twfe_l, adl_full, adl_all, adl_nofe
simulations/ dynamics/ results/ sim_direct_carryover_results.csv	beta2, rho_Z, twfe_s_bias, twfe_s_mcse, twfe_s_sd, twfe_s_rmse, twfe_l_bias, twfe_l_mcse, twfe_l_sd, twfe_l_rmse, adl_Ylag_bias, adl_Ylag_mcse, adl_Ylag_sd, adl_Ylag_rmse, adl_full_bias, adl_full_mcse, adl_full_sd, adl_full_rmse, adl_Dlag_bias, adl_Dlag_mcse, adl_Dlag_sd, adl_Dlag_rmse, adl_DYlag_bias, adl_DYlag_mcse, adl_DYlag_sd, adl_DYlag_rmse, adl_DZlag_bias, adl_DZlag_mcse, adl_DZlag_sd, adl_DZlag_rmse, adl_all_bias, adl_all_mcse, adl_all_sd, adl_all_rmse, adl_all_nofe_bias, adl_all_nofe_mcse, adl_all_nofe_sd, adl_all_nofe_rmse, b2_Dlag_mean, b2_DYlag_mean, b2_DZlag_mean, b2_all_mean, n_sims
simulations/ dynamics/ results/ sim_direct_carryover_raw.csv	twfe_s.D, twfe_l.D, adl_Ylag.D, adl_full.D, adl_Dlag.D, adl_DYlag.D, adl_DZlag.D, adl_all.D, adl_all_nofe.D, b2_Dlag.D_lag, b2_DYlag.D_lag, b2_DZlag.D_lag, b2_all.D_lag, sim, beta2, rho_Z
simulations/ dynamics/ results/ sim_direct_carryover_timing.csv	beta2, rho_Z, elapsed_s, total_s
simulations/ dynamics/ results/ sim_direct_feedback_results.csv	phi, rho_Z, twfe_s_bias, twfe_s_mcse, twfe_s_sd, twfe_s_rmse, twfe_l_bias, twfe_l_mcse, twfe_l_sd, twfe_l_rmse, adl_Ylag_bias, adl_Ylag_mcse, adl_Ylag_sd, adl_Ylag_rmse, adl_full_bias, adl_full_mcse, adl_full_sd, adl_full_rmse, pooled_l_bias, pooled_l_mcse, pooled_l_sd, pooled_l_rmse, adl_Dlag_bias, adl_Dlag_mcse, adl_Dlag_sd, adl_Dlag_rmse, n_sims
simulations/ dynamics/ results/ sim_direct_feedback_raw.csv	twfe_s.D, twfe_l.D, adl_Ylag.D, adl_full.D, pooled_l.D, adl_Dlag.D, sim, phi, rho_Z
simulations/ dynamics/ results/ sim_direct_feedback_timing.csv	phi, rho_Z, elapsed_s, total_s
simulations/ dual_role_z/ results/ sim_dual_role_z_8models_results.csv	rho_Z, sigma_aZ, pooled_s_bias, pooled_s_mcse, pooled_s_sd, pooled_l_bias, pooled_l_mcse, pooled_l_sd, twfe_s_bias, twfe_s_mcse, twfe_s_sd, twfe_l_bias, twfe_l_mcse, twfe_l_sd, adl_s_nofe_bias, adl_s_nofe_mcse, adl_s_nofe_sd, adl_l_nofe_bias, adl_l_nofe_mcse, adl_l_nofe_sd, adl_s_fe_bias, adl_s_fe_mcse, adl_s_fe_sd, adl_l_fe_bias, adl_l_fe_mcse, adl_l_fe_sd, n_sims
simulations/ dual_role_z/ results/ sim_dual_role_z_asymmetry_results.csv	pooled_s.D, pooled_l.D, twfe_s.D, twfe_l.D, adl_s_nofe.D, adl_l_nofe.D, adl_s_fe.D, adl_l_fe.D, sim, rho_Z, sigma_aZ
simulations/ dual_role_z/ results/ sim_dual_role_z_results.csv	gamma_D, delta_Y, twfe_s_bias, twfe_l_bias, adl_s_bias, adl_l_bias, twfe_s_mcse, twfe_l_mcse, adl_s_mcse, adl_l_mcse
simulations/ dual_role_z/ results/ sim_dual_role_z_results.csv	rho_Z, ar_order, sigma_aZ, twfe_bias_short, twfe_bias_long, twfe_ivb, adl_bias_short, adl_bias_long, adl_ivb, n_sims, twfe_net, adl_net, best_abs_bias, best_model
simulations/ dual_role_z/ results/ sim_dual_role_z_raw.csv	twfe_s.D, twfe_l.D, adl_s.D, adl_l.D, sim, rho_Z, ar_order, sigma_aZ
simulations/ dual_role_z/ results/ sim_dual_role_z_firewall_results.csv	rho_Z, rho_Y, twfe_s_bias, twfe_l_bias, adl_s_bias, adl_l_bias, twfe_s_mcse, twfe_l_mcse, adl_s_mcse, adl_l_mcse
simulations/ dual_role_z/ results/ sim_dual_role_z_varyT_8models_results.csv	TT, rho_Z, pooled_s_bias, pooled_s_mcse, pooled_l_bias, pooled_l_mcse, twfe_s_bias, twfe_s_mcse, twfe_l_bias, twfe_l_mcse, adl_s_nofe_bias, adl_s_nofe_mcse, adl_l_nofe_bias, adl_l_nofe_mcse, adl_s_fe_bias, adl_s_fe_mcse, adl_l_fe_bias, adl_l_fe_mcse, n_sims
simulations/ dynamics/ results/ sim_feedback_Y_to_D_results.csv	phi, rho_Z, twfe_s_bias, twfe_s_mcse, twfe_s_sd, twfe_s_rmse, twfe_l_bias, twfe_l_mcse, twfe_l_sd, twfe_l_rmse, adl_Ylag_bias, adl_Ylag_mcse, adl_Ylag_sd, adl_Ylag_rmse, adl_full_bias, adl_full_mcse, adl_full_sd, adl_full_rmse, adl_Dlag_bias, adl_Dlag_mcse, adl_Dlag_sd, adl_Dlag_rmse, adl_DYlag_bias, adl_DYlag_mcse, adl_DYlag_sd, adl_DYlag_rmse, adl_DZlag_bias, adl_DZlag_mcse, adl_DZlag_sd, adl_DZlag_rmse, adl_all_bias, adl_all_mcse, adl_all_sd, adl_all_rmse, adl_all_nofe_bias, adl_all_nofe_mcse, adl_all_nofe_sd, adl_all_nofe_rmse, mean_theta, mean_pi, mean_ivb, mean_ivb_formula, mean_se_short, mean_se_long, mean_se_adl_all, mean_abs_ivb_over_beta, mean_abs_ivb_over_se, n_sims, n_explosive, n_est_failed, n_discarded
simulations/ dynamics/ results/ sim_feedback_Y_to_D_raw.csv	twfe_s.D, twfe_l.D, adl_Ylag.D, adl_full.D, adl_Dlag.D, adl_DYlag.D, adl_DZlag.D, adl_all.D, adl_all_nofe.D, theta, pi_hat, se_short, se_long, se_adl_all, sim, phi, rho_Z
simulations/ dynamics/ results/ sim_feedback_Y_to_D_timing.csv	step, elapsed_sec, n_scenarios, n_reps, n_workers
simulations/ dynamics/ results/ sim_feedback_carryover_results.csv	phi, beta2, rho_Z, twfe_s_bias, twfe_s_mcse, twfe_s_sd, twfe_s_rmse, twfe_l_bias, twfe_l_mcse, twfe_l_sd, twfe_l_rmse, adl_Ylag_bias, adl_Ylag_mcse, adl_Ylag_sd, adl_Ylag_rmse, adl_full_bias, adl_full_mcse, adl_full_sd, adl_full_rmse, adl_Dlag_bias, adl_Dlag_mcse, adl_Dlag_sd, adl_Dlag_rmse, adl_DYlag_bias, adl_DYlag_mcse, adl_DYlag_sd, adl_DYlag_rmse, adl_DZlag_bias, adl_DZlag_mcse, adl_DZlag_sd, adl_DZlag_rmse, adl_all_bias, adl_all_mcse, adl_all_sd, adl_all_rmse, adl_all_nofe_bias, adl_all_nofe_mcse, adl_all_nofe_sd, adl_all_nofe_rmse, b2_all_mean, n_sims
simulations/ dynamics/ results/ sim_feedback_carryover_raw.csv	twfe_s.D, twfe_l.D, adl_Ylag.D, adl_full.D, adl_Dlag.D, adl_DYlag.D, adl_DZlag.D, adl_all.D, adl_all_nofe.D, b2_all.D_lag, sim, phi, beta2, rho_Z

Table 4: Complete CSV headers recorded in the provenance manifest. (continued)

Artifact path	Full CSV header
simulations/ dynamics/ results/ sim_feedback_carryover_timing.csv	phi, beta2, rho_Z, elapsed_s, total_s
simulations/ dynamics/ results/ sim_jensen_two_period_summary.csv	scenario, att_true, spec, mean_estimate, bias, rmse, coverage, mcse, n_sims
simulations/ dynamics/ results/ sim_jensen_two_period_raw.csv	spec, estimate, std_error, conf_low, conf_high, p_value, rep, scenario, att_true, treat_share, n_units, n_obs
simulations/ dynamics/ results/ sim_jensen_two_period_decomp.csv	scenario, mean_base, mean_tvc, mean_safe, mean_full, achieved_base_minus_safe, achieved_full_minus_safe, achieved_tvc_minus_base, mcse_base_minus_safe, mcse_full_minus_safe, mcse_tvc_minus_base, target_base_minus_safe, target_full_minus_safe, target_tvc_minus_base, gap_base_minus_safe, gap_full_minus_safe, gap_tvc_minus_base
simulations/ dynamics/ results/ sim_jensen_two_period_scenarios.csv	sigma_alpha, lambda_post, v_intercept, v_x1, v_x2, d_intercept, d_x1, d_x2, d_v, y_v, y_x1, y_x2, sigma_y0, sigma_y1, sigma_zm0, sigma_zm1, sigma_zc0, sigma_zc1, zm0_x1, zm0_x2, zc0_x1, zc0_x2, zm_ar, zc_ar, zm_x1, zm_x2, zc_x1, zc_x2, zm_u, zc_u, y_zm, y_zc, y_u, tau_direct, zm_d, zc_d, target_base_minus_safe, target_full_minus_safe, target_tvc_minus_base, scenario, att_true
simulations/ v4_mechanisms/ results/ sim_mechC_adl_results.csv	prob_switch, delta_D, delta_Y, rho_Z, twfe_s_bias, twfe_s_mcse, twfe_s_sd, twfe_s_rmse, twfe_l_bias, twfe_l_mcse, twfe_l_sd, twfe_l_rmse, adl_Ylag_bias, adl_Ylag_mcse, adl_Ylag_sd, adl_Ylag_rmse, adl_full_bias, adl_full_mcse, adl_full_sd, adl_full_rmse, adl_Dlag_bias, adl_Dlag_mcse, adl_Dlag_sd, adl_Dlag_rmse, adl_DYlag_bias, adl_DYlag_mcse, adl_DYlag_sd, adl_DYlag_rmse, adl_DZlag_bias, adl_DZlag_mcse, adl_DZlag_sd, adl_DZlag_rmse, adl_all_bias, adl_all_mcse, adl_all_sd, adl_all_rmse, adl_all_nofe_bias, adl_all_nofe_mcse, adl_all_nofe_sd, adl_all_nofe_rmse, mean_theta, mean_pi, mean_ivb, mean_ivb_formula, mean_se_short, mean_se_long, mean_se_adl_all, mean_abs_ivb_over_beta, mean_abs_ivb_over_se, n_sims, n_explosive, n_collinear, n_discarded
simulations/ v4_mechanisms/ results/ sim_mechC_adl_raw.csv	twfe_s.D, twfe_l.D, adl_Ylag.D, adl_full.D, adl_Dlag.D, adl_DYlag.D, adl_DZlag.D, adl_all.D, adl_all_nofe.D, theta, pi_hat, se_short, se_long, se_adl_all, sim, prob_switch, delta_D, delta_Y, rho_Z
simulations/ v4_mechanisms/ results/ sim_mechC_adl_timing.csv	step, elapsed_sec, n_scenarios, n_reps, n_workers
simulations/ nonlinearity/ results/ sim_nl_carryover_results.csv	carryover_type, beta_nl, rho_Z, twfe_s_bias, twfe_s_mcse, twfe_s_sd, twfe_s_rmse, twfe_l_bias, twfe_l_mcse, twfe_l_sd, twfe_l_rmse, adl_Ylag_bias, adl_Ylag_mcse, adl_Ylag_sd, adl_Ylag_rmse, adl_full_bias, adl_full_mcse, adl_full_sd, adl_full_rmse, adl_Dlag_bias, adl_Dlag_mcse, adl_Dlag_sd, adl_Dlag_rmse, adl_DYlag_bias, adl_DYlag_mcse, adl_DYlag_sd, adl_DYlag_rmse, adl_DZlag_bias, adl_DZlag_mcse, adl_DZlag_sd, adl_DZlag_rmse, adl_all_bias, adl_all_mcse, adl_all_sd, adl_all_rmse, adl_all_nofe_bias, adl_all_nofe_mcse, adl_all_nofe_sd, adl_all_nofe_rmse, n_sims, k, n_discarded
simulations/ nonlinearity/ results/ sim_nl_carryover_raw.csv	twfe_s.D, twfe_l.D, adl_Ylag.D, adl_full.D, adl_Dlag.D, adl_DYlag.D, adl_DZlag.D, adl_all.D, adl_all_nofe.D, sim, carryover_type, beta_nl, k, rho_Z
simulations/ nonlinearity/ results/ sim_nl_carryover_timing.csv	carryover_type, beta_nl, k, rho_Z, elapsed_s, n_discarded, total_s
simulations/ nonlinearity/ results/ sim_nl_collider_results.csv	nl_type, nl_strength, nl_Y, rho_Z, TT, twfe_s_bias, twfe_s_mcse, twfe_s_sd, twfe_s_rmse, twfe_l_bias, twfe_l_mcse, twfe_l_sd, twfe_l_rmse, adl_Ylag_bias, adl_Ylag_mcse, adl_Ylag_sd, adl_Ylag_rmse, adl_full_bias, adl_full_mcse, adl_full_sd, adl_full_rmse, adl_Dlag_bias, adl_Dlag_mcse, adl_Dlag_sd, adl_Dlag_rmse, adl_DYlag_bias, adl_DYlag_mcse, adl_DYlag_sd, adl_DYlag_rmse, adl_DZlag_bias, adl_DZlag_mcse, adl_DZlag_sd, adl_DZlag_rmse, adl_all_bias, adl_all_mcse, adl_all_sd, adl_all_rmse, adl_all_nofe_bias, adl_all_nofe_mcse, adl_all_nofe_sd, adl_all_nofe_rmse, n_sims, n_discarded
simulations/ nonlinearity/ results/ sim_nl_collider_raw.csv	twfe_s.D, twfe_l.D, adl_Ylag.D, adl_full.D, adl_Dlag.D, adl_DYlag.D, adl_DZlag.D, adl_all.D, adl_all_nofe.D, sim, nl_type, nl_strength, nl_Y, rho_Z, TT
simulations/ nonlinearity/ results/ sim_nl_collider_timing.csv	nl_type, nl_strength, nl_Y, rho_Z, TT, n_discarded, total_s
simulations/ nonlinearity/ results/ sim_nl_interact_results.csv	delta_DH, rho_Z, twfe_s_bias, twfe_s_mcse, twfe_s_sd, twfe_s_rmse, twfe_l_bias, twfe_l_mcse, twfe_l_sd, twfe_l_rmse, adl_Ylag_bias, adl_Ylag_mcse, adl_Ylag_sd, adl_Ylag_rmse, adl_full_bias, adl_full_mcse, adl_full_sd, adl_full_rmse, adl_Dlag_bias, adl_Dlag_mcse, adl_Dlag_sd, adl_Dlag_rmse, adl_DYlag_bias, adl_DYlag_mcse, adl_DYlag_sd, adl_DYlag_rmse, adl_DZlag_bias, adl_DZlag_mcse, adl_DZlag_sd, adl_DZlag_rmse, adl_all_bias, adl_all_mcse, adl_all_sd, adl_all_rmse, adl_all_nofe_bias, adl_all_nofe_mcse, adl_all_nofe_sd, adl_all_nofe_rmse, n_sims, n_discarded
simulations/ nonlinearity/ results/ sim_nl_interact_raw.csv	twfe_s.D, twfe_l.D, adl_Ylag.D, adl_full.D, adl_Dlag.D, adl_DYlag.D, adl_DZlag.D, adl_all.D, adl_all_nofe.D, sim, delta_DH, rho_Z
simulations/ nonlinearity/ results/ sim_nl_interact_timing.csv	delta_DH, rho_Z, elapsed_s, n_discarded, total_s
simulations/ nonlinearity/ results/ sim_overcontrol_contemporaneous_results.csv	nl_type, nl_strength, rho_Z, twfe_s_mean, twfe_s_mcse, twfe_bad_mean, twfe_bad_mcse, adl_Ylag_mean, adl_Ylag_mcse, adl_total_mean, adl_total_mcse, adl_bad_mean, adl_bad_mcse, adl_safe_mean, adl_safe_mcse, adl_both_mean, adl_both_mcse, ivb_twfe_diff_mean, ivb_twfe_formula_mean, ivb_twfe_formula_gap, ivb_adl_diff_mean, ivb_adl_formula_mean, ivb_adl_formula_gap, n_sims, twfe_s_bias_vs_total, twfe_bad_bias_vs_direct, adl_total_bias_vs_total, adl_bad_bias_vs_direct, adl_safe_bias_vs_total, overcontrol_twfe, overcontrol_adl, safe_shift_direct
simulations/ nonlinearity/ results/ sim_overcontrol_contemporaneous_raw.csv	twfe_s, twfe_bad, adl_Ylag, adl_total, adl_bad, adl_safe, adl_both, theta_twfe_bad, theta_adl_bad, pi_twfe, pi_adl, ivb_twfe_formula, ivb_adl_formula, ivb_twfe_diff, ivb_adl_diff, sim, nl_type, nl_strength, rho_Z
simulations/ nonlinearity/ results/ sim_overcontrol_contemporaneous_timing.csv	started_at, finished_at, elapsed_minutes

Table 4: Complete CSV headers recorded in the provenance manifest. (continued)

Artifact path	Full CSV header
simulations/ overcontrol/ results/ sim_overcontrol_results.csv	variant, rho_Z, nl_type, nl_strength, gamma_D, gamma_Y, twfe_s, twfe_Z, twfe_Zlag, adl_Ylag, adl_Ylag_Z, adl_Ylag_Zlag, adl_DYlag, adl_DYlag_Z, adl_all, adl_all_nofe, n_valid, n_discarded, twfe_s_bias_direct, twfe_Z_bias_direct, twfe_Zlag_bias_direct, adl_Ylag_bias_direct, adl_Ylag_Z_bias_direct, adl_Ylag_Zlag_bias_direct, adl_DYlag_bias_direct, adl_DYlag_Z_bias_direct, adl_all_bias_direct, adl_all_nofe_bias_direct, twfe_s_bias_total_linear, twfe_Z_bias_total_linear, twfe_Zlag_bias_total_linear, adl_Ylag_bias_total_linear, adl_Ylag_Zlag_bias_total_linear, adl_DYlag_bias_total_linear, adl_DYlag_Z_bias_total_linear, adl_all_bias_total_linear, adl_all_nofe_bias_total_linear, empirical_total, twfe_s_bias_empirical, twfe_Z_bias_empirical, twfe_Zlag_bias_empirical, adl_Ylag_bias_empirical, adl_Ylag_Z_bias_empirical, adl_Ylag_Zlag_bias_empirical, adl_DYlag_bias_empirical, adl_DYlag_Z_bias_empirical, adl_all_bias_empirical, adl_all_nofe_bias_empirical, ivb_twfe, ivb_adl_Ylag, ivb_adl_DYlag, expected_ivb_linear
simulations/ overcontrol/ results/ sim_overcontrol_raw.csv	twfe_s, twfe_Z, twfe_Zlag, adl_Ylag, adl_Ylag_Z, adl_Ylag_Zlag, adl_DYlag, adl_DYlag_Z, adl_all, adl_all_nofe, sim, variant, nl_type, nl_strength, rho_Z, gamma_D, gamma_Y, n_discarded, n_valid
simulations/ overcontrol/ results/ sim_overcontrol_grid.csv	variant_idx, nl_idx, rho_Z, variant, gamma_D, gamma_Y, nl_type, nl_strength, delta_D2, delta_log4, delta_softpoly2, delta_power15, delta_sin, delta_invlogit, delta_tanh, delta_Dlog
simulations/ dynamics/ results/ sim_persistent_confounder_results.csv	kappa, phi_U, rho_Z, delta_U, twfe_s_mean, twfe_s_bias, twfe_s_mcse, twfe_s_sd, twfe_l_mean, twfe_l_bias, twfe_l_mcse, twfe_l_sd, twfe_l_rmse, adl_Ylag_mean, adl_Ylag_bias, adl_Ylag_mcse, adl_Ylag_sd, adl_Ylag_rmse, adl_full_mean, adl_full_bias, adl_full_mcse, adl_full_sd, adl_full_rmse, adl_Dlag_mean, adl_Dlag_bias, adl_Dlag_mcse, adl_Dlag_sd, adl_Dlag_rmse, adl_DYlag_mean, adl_DYlag_bias, adl_DYlag_mcse, adl_DYlag_sd, adl_DYlag_rmse, adl_DZlag_mean, adl_DZlag_bias, adl_DZlag_mcse, adl_DZlag_sd, adl_DZlag_rmse, adl_all_mean, adl_all_bias, adl_all_mcse, adl_all_sd, adl_all_rmse, adl_all_nofe_mean, adl_all_nofe_bias, adl_all_nofe_mcse, adl_all_nofe_sd, adl_all_nofe_rmse, oracle_adl_mean, oracle_adl_bias, oracle_adl_mcse, oracle_adl_sd, oracle_adl_rmse, oracle_twfe_mean, oracle_twfe_bias, oracle_twfe_mcse, oracle_twfe_sd, oracle_twfe_rmse, lagid_mean, lagid_dev, lagid_mcse, lagid_sd, lagid_ctl_mean, lagid_ctl_dev, lagid_ctl_mcse, lagid_ctl_sd, n_sims, cor_DU_mean, n_discarded
simulations/ dynamics/ results/ sim_persistent_confounder_raw.csv	twfe_s.D, twfe_l.D, adl_Ylag.D, adl_full.D, adl_Dlag.D, adl_DYlag.D, adl_DZlag.D, adl_all.D, adl_all_nofe.D, oracle_adl.D, oracle_twfe.D, lagid.D_lag, lagid_ctl.D_lag, sim, cor_DU, kappa, phi_U, rho_Z, delta_U
simulations/ dynamics/ results/ sim_persistent_confounder_timing.csv	kappa, phi_U, rho_Z, delta_U, n_discarded, total_s
simulations/ dynamics/ results/ sim_staggered_posttreat_results.csv	prob_never, rho_Y, delta_Y, rho_Z, twfe_s_mean, twfe_s_bias, twfe_s_mcse, twfe_l_mean, twfe_l_bias, twfe_l_mcse, adl_Ylag_mean, adl_Ylag_bias, adl_Ylag_mcse, adl_full_mean, adl_full_bias, adl_full_mcse, adl_Dlag_mean, adl_Dlag_bias, adl_Dlag_mcse, adl_DYlag_mean, adl_DYlag_bias, adl_DYlag_mcse, adl_DZlag_mean, adl_DZlag_bias, adl_DZlag_mcse, adl_all_mean, adl_all_bias, adl_all_mcse, adl_all_nofe_mean, adl_all_nofe_bias, adl_all_nofe_mcse, n_sims
simulations/ dynamics/ results/ sim_staggered_posttreat_raw.csv	twfe_s.D, twfe_l.D, adl_Ylag.D, adl_full.D, adl_Dlag.D, adl_DYlag.D, adl_DZlag.D, adl_all.D, adl_all_nofe.D, sim, prob_never, rho_Y, delta_Y, rho_Z
simulations/ dynamics/ results/ sim_staggered_posttreat_timing.csv	prob_never, rho_Y, delta_Y, rho_Z, n_discarded, total_s
simulations/ v1_twfe/ sim_ivb_twfe_results.csv	gamma_Y, gamma_D, delta, R2_within, N, TT, beta, stable, mean_beta_short, mean_beta_long, mean_ivb, mean_ivb_formula, mean_abs_ivb_over_beta, mean_abs_ivb_over_se, mean_abs_ivb_over_sd_y, bias_short, bias_long, rmse_short, rmse_long, coverage_short, coverage_long, rejection_short, rejection_long
simulations/ v4_mechanisms/ results/ sim_ivb_twfe_v4_mechA.csv	mean_beta_short, mean_beta_long, mean_ivb, mean_ivb_formula, mean_theta, mean_pi, mean_abs_ivb_over_beta, mean_abs_ivb_over_se, mean_se_short, mean_se_long, bias_short, bias_long, rmse_short, rmse_long, coverage_short, coverage_long, mechanism, gamma_D_btw, gamma_D_wth, gamma_Y, R2_within, N, TT, beta
simulations/ v4_mechanisms/ results/ sim_ivb_twfe_v4_mechB.csv	mean_beta_short, mean_beta_long, mean_ivb, mean_ivb_formula, mean_theta, mean_pi, mean_abs_ivb_over_beta, mean_abs_ivb_over_se, mean_se_short, mean_se_long, bias_short, bias_long, rmse_short, rmse_long, coverage_short, coverage_long, mechanism, gamma_D, gamma_Y_btw, gamma_Y_wth, R2_within, N, TT, beta
simulations/ v4_mechanisms/ results/ sim_ivb_twfe_v4_mechC.csv	mean_beta_short, mean_beta_long, mean_ivb, mean_ivb_formula, mean_theta, mean_pi, mean_abs_ivb_over_beta, mean_abs_ivb_over_se, mean_se_short, mean_se_long, bias_short, bias_long, rmse_short, rmse_long, coverage_short, coverage_long, mechanism, gamma_D, gamma_Y, prob_switch, N, TT, beta
simulations/ v4_mechanisms/ results/ sim_ivb_twfe_v4_mechD.csv	mean_beta_short, mean_beta_long, mean_ivb, mean_ivb_formula, mean_theta, mean_pi, mean_abs_ivb_over_beta, mean_abs_ivb_over_se, mean_se_short, mean_se_long, bias_short, bias_long, rmse_short, rmse_long, coverage_short, coverage_long, mechanism, gamma_D, gamma_Y, sigma2_me, N, TT, beta

6 Static validation record

The builder checks the required inventory fields, unique identifiers, existence of every source script and declared result artifact, and records artifact size, header, and MD5. For numeric parameter columns present in artifacts, it checks probabilities/persistence terms on $[0,1]$ and positive count fields; for date-like columns it rejects unparsable dates and dates after the audit date. It does not execute simulation functions, therefore it makes no new Monte Carlo draws. The following gaps remain material:

1. `sim_ivb_completa.R` contains legacy exploratory simulations without a versioned persisted result file, so it cannot support a numerical manuscript claim.
2. Some scripts allow repetitions to be overridden by environment variables or command-line arguments. The source default is reported; actual execution settings should be recovered from the saved result/timing artifact before quoting a number.
3. The inventory maps source scripts to artifacts by their explicit output paths. It does not establish Git commit ancestry for historical result files; MD5 values make the audited artifact state reproducible.